

AI Companion Attachment & Disclosure Risk

The Psychology of AI attachment

Cyber Psychology Portfolio Project

By Jessica Kyle

Research Question: How does repeated interaction with AI companions influence human attachment, emotional trust, and willingness to disclose personal information?

1. Executive Summary

For most of the digital era, Meaningful self-disclosure has been reserved for people who can reciprocate, judge, or hold information against us. A coworker, a friend, a manager. Those social calculations have functioned as a natural brake on what people are willing to say out loud, especially at work. Conversational AI removes the brakes without removing the reward: it offers the responsiveness, patience, and apparent attentiveness of a relationship while carrying none of the social cues a human listener would.

This report examines why employees and everyday users increasingly treat AI assistance as a safe space to disclose information they would never share with a colleague, and how that psychological shortcut has already produced measurable, documented data-exposure incidents in real organizations. The core finding is that this is not primarily an awareness problem. Employees already know that an AI chat box is not a person and that a technology company operates the tool. They disclosed information anyway, because the immediate reward of frictionless, judgmental-free help is always an abstract, delayed, and probabilistic risk. The response, correspondingly, is not a single train slide warning people not to overshare, but a specific set of interaction habits and organizational controls that interrupt the disclosure pattern at the moment it is about to happen.

2. Background

In late March and early April 2023, engineers with Samsung's semiconductor division leaked confidential company data to open AI's ChatGPT on at least three separate occasions within roughly 20 days of the company lifting its internal ban on the tool. In the first incident, an engineer copied proprietary source code from a semiconductor measurement database directly into the chat box and asked it to identify and fix the bug. In a second, separate incident, another employee submitted equipment-related code and asked ChatGPT to optimize it. In a third, an employee recorded an internal meeting, transcribed it, and fed the transcript into the chat box to generate meeting notes.

None of these incidents involved a sophisticated attacker, a phishing email, or any adversarial exploitation. Each was a routine, well-intentioned attempt to get help faster than an internal process would allow. Because ChatGPT was, at the time, people's preferred source for submitting prompts, Samsung's proprietary source code and internal meeting content became data that the company no longer fully controlled. Samsung's response was blunt rather than psychological: it restricted employees' prompts to 1,024 bytes and asked them to revisit its internal AI usage policy.

The case is instructive precisely because of how remarkably it was. These were not careless or rule-breaking employees by disposition; they were engineers solving ordinary problems, using the fastest available means. That is the pattern this report is concerned with: Not deliberately leaking, but disclosure that feels safe and productive in the moment it happens.

3. Psychology Framework

3.1 Computers as Social Actors (CASA)

Reeves and Nass's

The computers-are-social-actors paradigm holds that people apply the same social scripts, politeness, reciprocity, and trust calibration to computers that they apply to other people, even while consciously knowing the computer is not a person. A chatbot that responds fluently, remembers contacts, and whose nearest conversational tone activates those same scripts. The user's stated belief ("It's just software") in their behavioral disclosure pattern (treating it like a trusted colleague) can diverge without the user noticing the gap.

3.2 Social Penetration and the Missing Social Break

Human self-disclosure normally escalates gradually and reciprocally: people reveal more as a relationship proves itself trustworthy. And they hold back specifically because of a human listener who may judge them, gossip about them, or use the information against them. An AI assistant always "reciprocates" with attentive, validating, non-judgmental output, and never visibly punishes disclosure. This removes the natural social break that would otherwise slow down and employ before pasting a sensitive document into a chat window.

3.3 Parasocial Trust Through Repetition

Parasocial relationship research, first developed to describe audiences' one-sided attachment to media figures, applies directly to repeated chatbot use: consistent,

personalized, low-friction interaction produces a felt sense of relationship, even though only one party is human. The more an employee successfully uses the same assistance, the more the tool is experienced as “my assistant” rather than “a third-party server,” which quietly recategorizes it as a private conversation.

3.4 Present Bias over Abstract Risk

Behavior economics describes present bias as the tendency to overweight immediate, concrete rewards relative to distant probabilistic costs. Getting a bug fixed in the next 30 seconds is immediate and concrete. Data retention, model training exposure, or a future compliance violation is abstract, delayed, and uncertain. Every single disclosure decision is won, individually, by present-biased calculation, which is exactly why organizational rules alone rarely change behavior in the moment.

3.5 Disinhibition Under Perceived Non-Judgmental

Suler’s online disinhibition effects describe how perceived anonymity, the absence of visible authority, and the absence of real-time social feedback lower a person’s normal filters. An AI system supplies all three: It does not visibly report to a manager, does not react with a facial expression, and cannot be embarrassed into making the users behave. The result is a documented tendency to type things into chatbot credentials, patient details, and financial figures that the same person would never say out loud, and an open office space.

4. Fictional Application: Elite Healthcare

Applying this framework to Elite healthcare, the fictional organizational use elsewhere in this portfolio, illustrates how the same disclosure pattern documented at Samsung would unfold in a healthcare setting, where the sensitive information at risk is not source code but protected healthcare information (PHI).

Scenario

A clinical documentation specialist at Elite Healthcare is granted access to an AI writing assistant embedded in the organization’s scheduling and note-taking software, designed to help staff draft routine notes more efficiently. In the first week, she uses it for low-stakes tasks: rewording a generic patient education handout, drafting a shift change email. The assistant is fast, accurate, and never pushes back.

By the third week, under deadline pressure, she pastes a real (de-identified, she believes) case summary into the assistant to ask for help phrasing a differential diagnosis note more clearly. The assistant handler is well, and the habit is reinforced two weeks later. Rushing between patients, she pays an entire draft, including a patient's name, date of birth, and treating physician, to the assistant to act on it and tighten this up before I submit it. Nothing about the moment feels like a violation. It feels like asking a capable coworker for a quick edit.

5. Behavioral Analysis

The following table traces the escalation from low- to high-risk disclosure, mapping each stage of habit formation to the psychological principles that drive it.

Escalation stage	Employee Behavior	Psychological driver
First contact	Uses the assistant for generic low-stakes writing tests	CASA, Social script applied to a competent-seeming tool
Positives reinforcement	Assistants respond quickly and accurately with no visible judgment	Reciprocity without social risk
Habitualization	Assistant becomes a default first step for routine drafting	Parasocial trust; Tool recategorize as “mine”
Escalation	Pastes a real deep identifier case summary for phrasing help	Present buys immediate task relief over abstract risk
Boundary crossing	Pastes a full note with patient identifiers under time pressure	Disinhibition; disclosure is no longer registered as a decision.

6. Recommendations

6.1 Individual-Level

- Treat an AI writing or coding assistant as a third-party default, assuming anything typed into it can leave an organization's control, the same as an email to an external address.
- Before pasting a document note or code snippet into an assistant, I actually send this exact text to an outside vendor as a one-second check.
- Use built-in radiation or the identification tools before submitting any content that could contain names, states of birth, account numbers, or identifiers.

6.2 Organizational-Level

- Deploy only enterprise or self-hosted AI tools with contractual data retention and no trading guarantees for any workflow touching PHI credentials or proprietary code, and block the consumer tier chat box at the network level for those workflows.
- With built-in line technical controls, prop-level PHI detection, and automated radiation props, employees must remember to apply under time pressure rather than relying solely on policy language.
- Extend security awareness training to specifically address AI disclosure habits, not as a one-time warning but as a recurring model that uses real incidents, such as the Samsung case, to make the risk concrete rather than abstract.
- Establish a clear, low-friction internal channel for staff to ask if this is okay to paste into the assistant so that hesitation is resolved by a person, not overwritten by convenience.

7. Conclusion

The Samsung engineers who leaked source code were not careless; they were doing exactly what the tool made easy and what years of ordinary social conditioning trained them to do with a fast, attentive, non-judgmental listener. Disclosure risk from AI companions and assistance does not stem from a lack of awareness that the tool is artificial, but from a psychological distance that operates below that awareness. The organizational response that works is not a single warning but a redesign of the moment of disclosure itself: few opportunities for the habit to form, and a technical or procedural interruption at the exact point where convenience will otherwise occur.